Chia-Kai Kuo

Postdoctoral Researcher, Max Planck Institute for Physics, Garching, Germany

chia-kai.kuo@mpp.mpg.de chiakaikuo@gmail.com
ORCID: 0000-0002-1523-8025 INSPIRE-HEP chiakaikuo.github.io

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PROFILE

I study the geometric and analytic structures of quantum field theory, focusing on scattering amplitudes, correlation functions, and cosmological wavefunction coefficients. My aim is to understand how these observables can be constructed from geometry, symmetry, and their singularities. My doctoral research focused on loop amplitudes in ABJM theory and their description in terms of positive geometry.

ACADEMIC POSITION

Postdoctoral Researcher, Max Planck Institute for Physics, Garching, Germany June 2024 – Present

EDUCATION

Ph.D. in Theoretical Physics, National Taiwan University, Taipei, Taiwan 01/02/2021 – 30/05/2024

Supervisors: Song He and Yu-tin Huang

Thesis: *Loop Amplitudes of ABJM and Its Positive Geometry*

M.Sc. in Physics, National Taiwan University, Taipei, Taiwan 01/09/2018 – 31/08/2020

B.Sc. in Physics, National Taiwan University, Taipei, Taiwan 01/09/2014 – 31/08/2018

PUBLICATIONS

10 peer-reviewed articles and 1 preprint. For the current publication record and citation data, see my INSPIRE-HEP profile.

Refereed Publications

1. C.-K. Kuo and Q. Yang, "Notes on off-shell conformal integrals and correlation functions at five points", Phys. Rev. D 113 (2026) 105022. arXiv:2512.21947.

   Description: We construct pure two-loop five-point off-shell conformal integral bases and use them to compute five-point half-BPS correlators at symbol level, including both maximal and non-maximal sectors.

2. S. He, Y.-T. Huang and C.-K. Kuo, "Leading singularities and chambers of Correlahedron", JHEP 03 (2026) 071. arXiv:2505.09808.

   Description: We relate chamber forms to local four-point correlator integrands through four loops, including an elliptic contribution, and use this organization to guide the choice of pure integral bases.

3. Y.-T. Huang, C.-K. Kuo and C. Zhang, "BDS ansatz in ABJM via scaffolding triangulations", JHEP 02 (2026) 022. arXiv:2510.20663.

   Description: We use scaffolding triangulations to organize the two-loop BDS integrand and prove elliptic-cut cancellation and triangulation independence of the integrated result at arbitrary even multiplicity.

4. S. He, Y.-T. Huang and C.-K. Kuo, "All-loop geometry for four-point correlation functions", Phys. Rev. D 110 (2024) L081701. arXiv:2405.20292.

   Description: We propose an all-loop positive geometry for four-point correlators in planar $\mathcal{N} = 4$ SYM and verify that its canonical forms reproduce known correlator integrands up to three loops.

5. S. He, Y.-T. Huang and C.-K. Kuo, "The ABJM Amplituhedron", JHEP 09 (2023) 165. arXiv:2306.00951.

   Erratum: JHEP 04 (2024) 064.

   Description: We develop a positive-geometric construction for planar ABJM loop integrands, obtaining explicit one-loop integrands through ten points and two-loop integrands through eight points.

6. S. He, C.-K. Kuo, Z. Li and Y.-Q. Zhang, "Emergent unitarity, all-loop cuts and integrations from the ABJM amplituhedron", JHEP 07 (2023) 212. arXiv:2303.03035.

   Description: We prove all-loop unitarity properties of the four-point ABJM amplituhedron and integrate its negative geometries up to two loops to extract infrared-finite data.

7. S. He, Y.-T. Huang, C.-K. Kuo and Z. Li, "The two-loop eight-point amplitude in ABJM theory", JHEP 02 (2023) 065. arXiv:2211.01792.

   Description: We compute the two-loop eight-point ABJM amplitude, fixing the integrand from dual conformal symmetry, maximal cuts and soft-collinear constraints, and analyse its integrated finite part.

8. S. He, C.-K. Kuo, Z. Li and Y.-Q. Zhang, "All-Loop Four-Point Aharony-Bergman-Jafferis-Maldacena Amplitudes from Dimensional Reduction of the Amplituhedron", Phys. Rev. Lett. 129 (2022) 221604. arXiv:2204.08297.

   Description: We define a reduced amplituhedron for four-point ABJM amplitudes by dimensional reduction of the $\mathcal{N} = 4$ SYM amplituhedron and obtain explicit integrands up to five loops.

9. S. He, C.-K. Kuo and Y.-Q. Zhang, "The momentum amplituhedron of SYM and ABJM from twistor-string maps", JHEP 02 (2022) 148. arXiv:2111.02576.

   Description: We relate twistor-string maps to momentum amplituhedra in $\mathcal{N} = 4$ SYM and ABJM, showing how tree amplitudes arise as canonical forms of positive geometries.

10. Y.-T. Huang, C.-K. Kuo and C. Wen, "Dualities for Ising networks", Phys. Rev. Lett. 121 (2018) 251604. arXiv:1809.01231.

    Description: We establish a correspondence between planar Ising networks and cells of the positive orthogonal Grassmannian, leading to recursive methods for computing network correlators.

Preprint

1. Y.-T. Huang, C.-K. Kuo, Y. Liu and J. Mei, "Beyond Discontinuities: Cosmological WFCs and the Supersymmetric Orthogonal Grassmannian", Preprint. arXiv:2604.08512.

   Description: We combine an inversion formula with supersymmetric Grassmannian representations to reconstruct tree-level wavefunction coefficients from energy discontinuities, with explicit three- and four-point results.

AWARDS AND HONORS

- **NTU-CTP Best Student Paper Award** 2023
- **NCTS Student Outstanding Paper Award** 2023

SELECTED TALKS AND SEMINARS

- **Beyond Discontinuities: Cosmological wavefunctions and the orthogonal Grassmannian**, Pollica, Italy 18/09/2026
- **Positive Geometry and Leading Singularities in Correlation Functions**, New Ways to Higher Points, Humboldt University of Berlin, Germany 01/10/2025
- **BDS ansatz in ABJM via scaffolding triangulations**, 42nd International Conference on High Energy Physics, Prague, Czech Republic. Conference; related paper. July 2024
- **Geometry of the Four-Point Correlators**, Kick-off Workshop on Positive Geometry in Particle Physics and Cosmology, Max Planck Institute for Mathematics in the Sciences, Leipzig, Germany 16/02/2024
- **The ABJM Amplituhedron**, Amplitudes 2023 Conference, CERN, Switzerland 07/08/2023
- **All-Loop Four-Point ABJM Amplitudes from Dimensional Reduction of the Amplituhedron**, 2022 NTU-Kyoto High Energy Physics Workshop / Kawai Fest, National Taiwan University, Taiwan 18/12/2022
- **On-shell method on ABJM two-loop amplitudes**, NTU-NCTS String Seminar, National Taiwan University, Taiwan 25/11/2022

TEACHING EXPERIENCE

Teaching Assistant for "Electricity and Magnetism", "Quantum Mechanics", and "Applied Mathematics I" at the Department of Physics, National Taiwan University.

RESEARCH TOOLS

Mathematica, Maple, and LaTeX.